Galvanic Cells and Electrode Potentials

Answer key with worked solutions

A. Understanding

1. Galvanic cell.

> **Oxidation** happens at the **anode** (negative) and **reduction** at the **cathode** (positive). Electrons flow through the wire **from anode to cathode**. The **salt bridge** lets ions move between the half-cells so that the solutions stay electrically neutral and the current can continue.

B. Cell potentials

2. Zinc–silver cell.

> (a) Ag^+/Ag has the more positive E°, so silver is the **cathode**; zinc is the **anode**. (b) E°cell = +0.80 V − (−0.76 V) = **+1.56 V** (c) $Zn + 2\,Ag^+ \longrightarrow Zn^{2+} + 2\,Ag$ (d) $Zn\,(s) \mid Zn^{2+}\,(aq) \,||\, Ag^+\,(aq) \mid Ag\,(s)$

3. Nickel–silver cell.

> E°cell = +0.80 V − (−0.25 V) = **+1.05 V**; **nickel** is the anode.

4. Magnesium–copper cell.

> E°cell = +0.34 V − (−2.37 V) = **+2.71 V** (magnesium is the anode).

5. Spontaneous?

> (a) Zn^{2+} would be reduced (cathode) and Cu oxidized (anode): E°cell = −0.76 V − (+0.34 V) = **−1.10 V**, so **not spontaneous**. (b) H^+ reduced, Zn oxidized: E°cell = 0.00 V − (−0.76 V) = **+0.76 V**, so **spontaneous** (zinc dissolves in acid).

6. Strongest agents.

> Strongest oxidizing agent: **F_2** (most positive E°, +2.87 V). Strongest reducing agent: **Mg** (most negative E°, −2.37 V, so Mg is most easily oxidized).

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C. Free energy

7. ΔG° for the Daniell cell.

> E°cell = +0.34 V − (−0.76 V) = +1.10 V; n = 2 mol of electrons.
>
> $$\Delta G^\circ = -(2\ \text{mol})\left(96\,485\ \tfrac{\text{C}}{\text{mol}}\right)(1.10\ \text{V})$$
> $$= -2.12 \times 10^5\ \text{J} = -212\ \text{kJ}$$
>
> (mol cancels, and C × V = J.)

D. Electrolysis

8. Silver plating.

> $t = 45.0\ \text{min} \times 60\ \text{s/min} = 2700\ \text{s}$; $Q = 1.50\ \text{C/s} \times 2700\ \text{s} = 4050\ \text{C}$
>
> $n(e^-) = \dfrac{4050\ \text{C}}{96\,485\ \text{C/mol}} = 0.04198\ \text{mol}$; 1 mol Ag per mol e⁻, so $n(\text{Ag}) = 0.04198\ \text{mol}$
>
> $m = 0.04198\ \text{mol} \times 107.87\ \text{g/mol} = \mathbf{4.53\ g}$

E. Challenge

9. Time to deposit 5.00 g Cu.

> $n(\text{Cu}) = \dfrac{5.00\ \text{g}}{63.55\ \text{g/mol}} = 0.07868\ \text{mol}$; $n(e^-) = 2 \times 0.07868\ \text{mol} = 0.1574\ \text{mol}$
>
> $Q = 0.1574\ \text{mol} \times 96\,485\ \text{C/mol} = 1.518 \times 10^4\ \text{C}$
>
> $t = \dfrac{Q}{I} = \dfrac{1.518 \times 10^4\ \text{C}}{3.00\ \text{C/s}} = 5.06 \times 10^3\ \text{s}$, and $5.06 \times 10^3\ \text{s} \times \dfrac{1\ \text{min}}{60\ \text{s}} = \mathbf{84.3\ min}$

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